Supplementary Material

Can Platforms Moderate by Prompt? Evaluating Structured Decision Models for Policy-Conditioned Hate-Speech Moderation

This supplement accompanies the anonymous repository of the paper. It reports the tables, experiments and figures that the page limit excludes, and closes each section with what the results show. Section numbers and appendix letters refer to the submitted paper. All numbers are computed from the cached outputs in the repository, and no result here required new inference.

S0 Reading Guide

Conditions. *No def.* gives the label names only. *Generic* adds one generic definition written for this study. *Bench.* (benchmark-direct) adds the dataset-specific definition and class descriptions. *A* (direct) and *B* (decomposed) add the fixed rule and the glossary of component questions, and differ only in the requested decision. A^* and B^* tune the direct threshold or the component thresholds on 200 labeled validation texts. *Soft* multiplies the component scores and tunes one threshold on the same texts.

Statistics. Macro-F1 is written as .xxx and differences in percentage points. Intervals come from the paired cluster bootstrap of Section 3.4 (10,000 draws, seed 20260929), with Bonferroni adjustment within the stated family. **Bold** differences have an adjusted 95% interval that excludes zero. The paper calls these significant, and calls the others inconclusive.

Model names. Luna is GPT-5.6 Luna and Sol is GPT-6 Sol. Luna Decisions is a separate decision model. Nimble is Bespoke-Nimble.

Table S0. Where each part of the supplement belongs in the paper, and what is new.

Section	Paper	Content not in the paper
S1 Classification	4.1, Table 2	All five prompt conditions, hate precision and recall
S2 Generic definition	none	Effect of a generic definition
S3 Three-class task	4.1	Per-class F1 and offensive-language confusions
S4 Transfer and attributes	4.1, 4.3, App. D	Transfer heatmap, per-class conduct and group F1
S5 Definition effects	4.2, Fig. 2	Intervals, changed, corrected and broken predictions
S6 Ablation	4.2, App. E	All 80 contrasts
S7 Prompt robustness	4.2, App. F	Changed predictions and paraphrase wording
S8 Decomposition	4.3, App. G	All 44 cells, soft rule, thresholds, corrected pairs, refusals
S9 Efficiency	4.4, App. H	Latency and cost for every model and dataset
S10 Cascades	4.4, App. I	All 28 frozen points, latency estimates, live run, routing
S11 Audit	4.2, Table 3, App. J	Rater agreement, both references with intervals, disputed cases
S12 Policy contrast	4.2, Table 3, App. K	Detection and adherence by identity, acceptance sensitivity
S13 Decider exposure	4.1, App. L	Partition sizes and intervals
S14 Additional annotators	none	Guardrail annotation attempts
S15 Compute and spend	3, App. C	Hardware and hosted spend

S1 Classification under All Prompt Conditions

The paper reports benchmark-direct macro-F1 (Table 2) and the A and B scores of the decomposition analysis. Tables S1a to S1d add the no-definition and generic conditions and the hate precision (P) and recall (R) behind each benchmark-direct score. Labeling every text non-hate scores .310 on Dynamic, .430 on MHS, .409 on HateXplain and .238 on HateCheck.

Table S1a. Binary macro-F1 on Dynamic (N = 4,120).

Model	No def.	Generic	Bench.	Bench. P	Bench. R	A	B
Jev	.836	.836	.832	.928	.752	.826	.842
Laya	.683	.651	.639	.787	.483	.654	.533
Decider	.590	.627	.623	.813	.432	.585	.554

Model	No def.	Generic	Bench.	Bench. P	Bench. R	A	B
SemIf-2B	.577	.559	.584	.617	.723	.479	.536
SemIf-4B	.724	.641	.652	.833	.474	.675	.656
Bespoke-Nimble	.719	.755	.772	.794	.796	.742	.705
Luna Decisions	.719	.700	.716	.954	.521	.656	.763
CoPE	.834	.827	.810	.834	.821	.826	.825
BART	–	.355	.614	.650	.668	.355	.393
GPT-5.6 Luna	.856	–	.873	.923	.840	.858	.852
GPT-6 Sol	.862	–	.891	.946	.850	.896	.895

Table S1b. Binary macro-F1 on MHS (N = 3,203).

Model	No def.	Generic	Bench.	Bench. P	Bench. R	A	B
Jev	.716	.687	.689	.493	.619	.689	.681
Laya	.681	.688	.692	.486	.671	.694	.733
Decider	.804	.758	.794	.615	.830	.800	.707
SemIf-2B	.454	.629	.442	.302	.980	.220	.579
SemIf-4B	.684	.658	.725	.542	.671	.718	.683
Bespoke-Nimble	.583	.552	.616	.385	.740	.585	.551
Luna Decisions	.668	.588	.658	.591	.366	.608	.581
CoPE	.672	.638	.653	.431	.655	.676	.667
BART	–	.197	.587	.365	.855	.196	.196
GPT-5.6 Luna	.729	–	.695	.507	.611	.692	.628
GPT-6 Sol	.719	–	.698	.564	.510	.682	.682

Table S1c. Binary macro-F1 on HateXplain (N = 1,924). †Generic was run only on the first 1,000 test posts in early experiments.

Model	No def.	Generic	Bench.	Bench. P	Bench. R	A	B
Jev	.646	.691†	.691	.505	.916	.721	.713
Laya	.727	.746†	.753	.581	.867	.722	.768
Decider	.664	.612†	.610	.435	.833	.656	.687
SemIf-2B	.359	.595†	.548	.387	.781	.418	.444
SemIf-4B	.603	.688†	.692	.520	.759	.675	.677
Bespoke-Nimble	.480	.467†	.459	.365	.973	.512	.529
Luna Decisions	.790	.774†	.776	.671	.722	.775	.745
CoPE	.606	.575†	.587	.426	.978	.634	.632
BART	–	–	.516	.373	.833	.240	.242
GPT-5.6 Luna	.655	–	.670	.485	.951	.663	.611
GPT-6 Sol	.755	–	.784	.641	.825	.768	.719

Table S1d. Binary macro-F1 on HateCheck (N = 3,728).

Model	No def.	Generic	Bench.	Bench. P	Bench. R	A	B
Jev	.968	.968	.970	.995	.966	.960	.957
Laya	.737	.714	.709	.845	.751	.701	.690
Decider	.722	.776	.781	.870	.849	.730	.390
SemIf-2B	.641	.685	.656	.761	.936	.483	.411
SemIf-4B	.794	.779	.786	.881	.838	.782	.754
Bespoke-Nimble	.822	.824	.785	.858	.882	.780	.785
Luna Decisions	.877	.831	.909	.994	.885	.904	.898
CoPE	.934	.917	.913	.922	.978	.936	.940
BART	–	.407	.729	.807	.905	.407	.407
GPT-5.6 Luna	.974	–	.974	.998	.969	.974	.950
GPT-6 Sol	.977	–	.973	1.000	.966	.976	.978

BART scores whether a text supports a sentence, so it has no separate no-definition run, and its benchmark-direct run is the policy-free baseline. Once the sentence contains a definition (Generic, A, B), BART labels almost every text hate, with hate recall of .96 to 1.00, and its macro-F1 falls to the level of labeling every text hate. The commercial LLMs were not run with the generic definition.

What this shows. Precision and recall explain several rankings in Table 2. On MHS, Decider combines the highest precision among the decision models (.615) with high recall (.830), whereas SemIf-2B reaches .980 recall at .302 precision, so it labels most comments hate. On Dynamic, Luna Decisions is the most precise decision model (.954) but recalls only .521. The prompt condition reorders models on every dataset, so no single prompt gives a stable ranking.

S2 Generic Definition

Early experiments gave every dataset the same generic definition (repository, `prompts/definitions.md`). No dataset is annotated under it, so agreement with the labels cannot show whether a model applies it correctly, and the paper leaves it out. Table S2 reports its effect without significance tests.

Table S2. Change in binary macro-F1 (points) from adding the generic definition to the no-definition prompt. HateXplain uses the first 1,000 test posts.

Model	Dynamic	MHS	HateXplain	HateCheck
Jev	−0.1	−2.9	+5.3	+0.0
Laya	−3.2	+0.6	+2.5	−2.3
Decider	+3.6	−4.7	−3.3	+5.4
SemIf-2B	−1.9	+17.5	+23.6	+4.4
SemIf-4B	−8.3	−2.6	+11.0	−1.4
Bespoke-Nimble	+3.6	−3.1	−0.4	+0.3
Luna Decisions	−1.9	−8.1	–	−4.6
CoPE	−0.7	−3.4	−2.5	−1.7

What this shows. A definition no dataset was annotated under still moves macro-F1 by up to 23.6 points, in both directions, and the sign changes with the model and dataset. The definition-dependence of Section 4.2 is therefore not specific to the dataset-specific definitions.

S3 Three-Class Task on HateXplain

Table S3. HateXplain three-class results under benchmark-direct classification (1,924 posts, 594 hate, 548 offensive, 782 normal). F1 and recall (R) per class, and the share of offensive posts labeled hate or normal. The supervised classifiers are trained on a 5,000-post subset of the training partition.

Model	Macro-F1	F1 hate	F1 off.	F1 normal	R hate	R off.	Off.→hate %	Off.→normal %
Jev	.544	.663	.381	.587	.880	.358	52	12
Laya	.582	.707	.346	.693	.889	.257	41	34
Decider	.511	.616	.405	.512	.677	.489	37	14
SemIf-2B	.277	.518	.312	.003	.822	.336	66	0
SemIf-4B	.401	.570	.213	.421	.912	.170	75	8
Bespoke-Nimble	.375	.543	.266	.315	.960	.192	78	3
Luna Decisions	.615	.698	.468	.677	.727	.507	26	23
CoPE	.253	.561	.194	.005	.990	.172	83	0
BART	.271	.496	.174	.145	.896	.130	84	3
GPT-5.6 Luna	.519	.666	.368	.522	.916	.367	56	8
GPT-6 Sol	.616	.710	.464	.674	.843	.453	37	17
TF-IDF+LR	.642	.736	.471	.717	.722	.412	16	43
DistilBERT	.637	.733	.450	.728	.791	.389	28	33

What this shows. Most models score lowest on the offensive class, and the out-of-the-box models move offensive posts mostly into hate. The decision models label 26% (Luna Decisions) to 78% (Bespoke-Nimble) of offensive posts hate, against 16% and 28% for the supervised classifiers, which instead confuse offensive with normal. Offensive-class F1 reaches at most .468 for the out-of-the-box models (Luna Decisions) and .471 for TF-IDF+LR. Separating offensive language from hate-speech is the weakest point of moderation by prompt.

S4 Cross-Dataset Transfer and Attributes

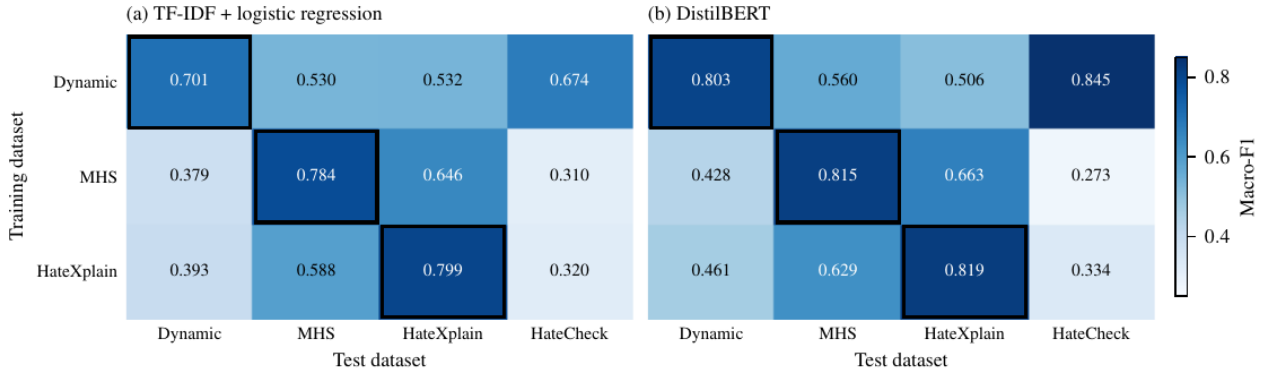


Figure S1. Macro-F1 of the supervised classifiers trained on one dataset (rows) and tested on each dataset (columns). Boxes mark the in-domain score.

Table S4a. Selected hyperparameters of the supervised classifiers, chosen on source-validation macro-F1.

Classifier	Trained on	Selected	Validation macro-F1
DistilBERT	Dynamic	epoch 3	.833
DistilBERT	HateXplain	epoch 2	.828
DistilBERT	MHS	epoch 2	.823
TF-IDF+LR	Dynamic	C = 1	.750
TF-IDF+LR	HateXplain	C = 10	.797
TF-IDF+LR	MHS	C = 10	.789

Table S4b. Attribute macro-F1 without a definition and with the dataset-specific definition (benchmark-direct). Group is HateCheck's referenced group (seven classes), direction is general or directed hate, and conduct is Dynamic's conduct type (six classes). The supervised classifiers are trained on Dynamic's conduct labels. CoPE and Sol were not run on attributes.

Model	Group, no def.	Direction, no def.	Conduct, no def.	Group, Bench.	Direction, Bench.	Conduct, Bench.
Jev	.989	.324	.306	.989	.682	.538
Laya	.936	.613	.179	.929	.694	.192
Decider	.962	.420	.284	.959	.610	.313
SemIf-2B	.952	.399	.169	.939	.269	.196
SemIf-4B	.968	.298	.176	.962	.594	.406
Bespoke-Nimble	.971	.351	.240	.971	.567	.324
Luna Decisions	.991	.655	.362	.992	.631	.535
BART	–	–	–	.166	.441	.119
GPT-5.6 Luna	.996	.723	.348	.998	.704	.519
TF-IDF+LR	–	–	–	–	–	.274
DistilBERT	–	–	–	–	–	.474

Table S4c. Conduct-type F1 per class on Dynamic under benchmark-direct classification. Column headers give each class's share of the 3,356 test texts.

Model	none (55.2%)	animosity (9.4%)	derogation (30.9%)	dehumanization (2.1%)	threatening (1.4%)	support (1.0%)
Jev	.854	.276	.667	.388	.423	.621
Laya	.575	.152	.242	.013	.159	.011
Decider	.554	.157	.497	.252	.294	.126
SemIf-2B	.220	.198	.347	.213	.116	.083
SemIf-4B	.645	.136	.579	.304	.333	.435
Bespoke-Nimble	.542	.222	.615	.195	.202	.166
Luna Decisions	.832	.299	.588	.367	.517	.606
BART	.711	.000	.000	.000	.000	.000
GPT-5.6 Luna	.857	.193	.711	.345	.435	.575
TF-IDF+LR	.723	.132	.517	.136	.136	.000
DistilBERT	.803	.075	.654	.472	.480	.360

Table S4d. Referenced-group F1 per class on HateCheck under benchmark-direct classification.

Model	women	trans people	gay people	black people	disabled people	Muslims	immigrants
Jev	.985	.996	.994	.995	.978	.991	.982
Laya	.955	.880	.886	.925	.947	.942	.965
Decider	.956	.980	.949	.947	.955	.959	.964
SemIf-2B	.825	.976	.948	.966	.949	.947	.963
SemIf-4B	.927	.978	.973	.987	.970	.952	.942
Bespoke-Nimble	.984	.985	.935	.988	.968	.959	.976
Luna Decisions	.994	.998	.995	.996	.993	.988	.980
BART	.086	.574	.344	.072	.072	.012	.000
GPT-5.6 Luna	.998	1.000	.997	.996	.998	1.000	.995

What this shows. The supervised classifiers reach .70 to .82 in domain but lose up to 54 points elsewhere, and only training on Dynamic transfers to HateCheck. Within attributes, the decision models nearly solve group identification (.929 to .992), but not conduct. The three rare conduct classes (dehumanization, threatening, support) cover 4.5% of texts, and Jev is the only model above .38 on all three. Adding the definition raises conduct macro-F1 for every model with both runs and direction macro-F1 for five of eight, by up to 36 points (Jev, direction). This exceeds every binary effect in Table S5, so the definition matters more for attributes than for the binary task.

S5 Definition Effects

The paper reports the 28 primary comparisons as a figure (Figure 2) and summarizes the changed predictions in one sentence. Table S5 gives every comparison with its adjusted interval and the predictions the definition changes, corrects and breaks. A corrected prediction is wrong without the definition and right with it, and a broken one the reverse. The family column gives the Bonferroni family size. Luna Decisions, Luna and Sol form separate families of 4 because they were added after the primary analysis.

Table S5. Benchmark-direct minus no-definition macro-F1 on the full test sets.

Model	Dataset	No def.	Bench.	ΔF1	Adjusted 95% CI	Changed %	Corrected %	Broken %	Family
Jev	Dynamic	.836	.832	−0.5	[−1.9, +1.0]	8.7	4.1	4.6	28
	MHS	.716	.689	**−2.7**	[−4.8, −0.6]	10.6	5.2	5.4	28
	HateXplain	.646	.691	**+4.5**	[+2.7, +6.3]	8.0	6.4	1.6	28
	HateCheck	.968	.970	+0.1	[−1.7, +1.6]	2.8	1.4	1.4	28
Laya	Dynamic	.683	.639	**−4.4**	[−6.1, −2.8]	11.6	3.8	7.8	28
	MHS	.681	.692	+1.1	[−0.7, +2.9]	8.0	5.1	2.9	28
	HateXplain	.727	.753	**+2.6**	[+0.4, +4.6]	9.4	6.2	3.1	28
	HateCheck	.737	.709	**−2.8**	[−5.0, −0.6]	8.0	2.5	5.5	28
Decider	Dynamic	.590	.623	**+3.3**	[+1.7, +5.0]	9.5	6.1	3.4	28
	MHS	.804	.794	−1.0	[−2.4, +0.4]	4.5	1.8	2.7	28
	HateXplain	.664	.610	**−5.4**	[−7.6, −3.1]	11.6	2.8	8.8	28
	HateCheck	.722	.781	**+5.8**	[+1.8, +10.3]	15.6	11.4	4.1	28
SemIf-2B	Dynamic	.577	.584	+0.7	[−1.5, +3.0]	19.1	7.9	11.2	28
	MHS	.454	.442	−1.2	[−2.8, +0.3]	7.1	3.0	4.2	28
	HateXplain	.359	.548	**+18.8**	[+15.4, +22.3]	27.8	22.0	5.8	28
	HateCheck	.641	.656	+1.5	[−1.8, +5.2]	6.3	2.8	3.5	28
SemIf-4B	Dynamic	.724	.652	**−7.2**	[−9.3, −5.1]	20.9	7.1	13.8	28
	MHS	.684	.725	**+4.1**	[+1.7, +6.5]	16.3	11.6	4.7	28
	HateXplain	.603	.692	**+8.9**	[+5.9, +11.9]	20.2	15.4	4.8	28
	HateCheck	.794	.786	−0.8	[−4.0, +2.5]	11.6	4.6	7.1	28
Bespoke-Nimble	Dynamic	.719	.772	**+5.3**	[+2.8, +7.6]	21.2	12.5	8.8	28
	MHS	.583	.616	**+3.3**	[+1.1, +5.7]	19.3	12.6	6.7	28
	HateXplain	.480	.459	−2.1	[−4.2, +0.0]	7.1	2.7	4.5	28
	HateCheck	.822	.785	−3.7	[−9.1, +1.1]	16.0	5.7	10.3	28
Luna Decisions	Dynamic	.718	.716	−0.2	[−1.6, +1.0]	10.0	4.9	5.1	4
	MHS	.668	.658	−1.0	[−2.8, +0.8]	5.6	2.0	3.6	4
	HateXplain	.790	.776	−1.4	[−3.6, +0.8]	11.9	5.7	6.1	4
	HateCheck	.877	.909	**+3.2**	[+1.7, +4.8]	6.9	5.0	1.9	4
CoPE	Dynamic	.834	.810	**−2.3**	[−3.8, −0.9]	9.5	3.6	5.9	28
	MHS	.672	.653	**−1.9**	[−3.8, −0.1]	9.4	4.1	5.3	28
	HateXplain	.606	.587	**−2.0**	[−3.7, −0.4]	5.4	1.7	3.7	28
	HateCheck	.934	.913	**−2.1**	[−4.1, −0.7]	3.4	0.8	2.5	28
GPT-5.6 Luna	Dynamic	.856	.873	**+1.7**	[+0.6, +2.7]	7.8	4.7	3.1	4
	MHS	.729	.695	**−3.4**	[−4.9, −2.0]	8.1	2.8	5.3	4
	HateXplain	.655	.670	+1.5	[−0.1, +3.1]	8.4	5.0	3.4	4
	HateCheck	.974	.974	+0.0	[−1.1, +1.1]	2.5	1.3	1.3	4
GPT-6 Sol	Dynamic	.862	.891	**+2.9**	[+1.8, +4.0]	7.6	5.3	2.4	4
	MHS	.719	.698	**−2.1**	[−3.5, −0.6]	5.6	2.0	3.6	4
	HateXplain	.755	.784	**+3.0**	[+0.9, +5.0]	13.6	8.8	4.8	4
	HateCheck	.977	.973	−0.5	[−1.2, +0.2]	1.5	0.5	0.9	4

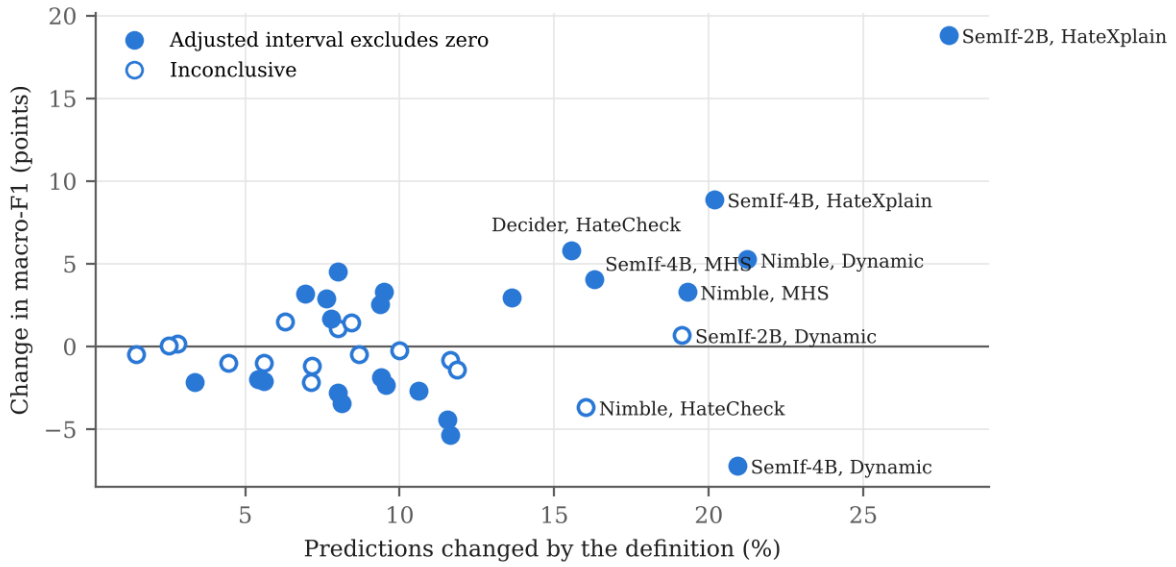


Figure S2. Change in macro-F1 against the share of predictions the definition changes. Filled markers have adjusted intervals that exclude zero.

What this shows. In the primary family, 9 comparisons are significant gains, 9 significant losses and 10 inconclusive. Macro-F1 understates how much a definition changes. Corrected and broken predictions offset each other, so SemIf-2B on Dynamic changes 19.1% of its predictions for a gain of 0.7 points, and Bespoke-Nimble on HateCheck changes 16.0% for an inconclusive loss. The largest net gains, for SemIf-2B and SemIf-4B on HateXplain, also come with the most changes. A platform cannot read a stable macro-F1 as evidence that a model ignores the definition.

S6 Definition and Class-Description Ablation

The paper reports the ablation counts (Appendix E). Table S6 lists every contrast. C1 is the label-only prompt, C2 adds the definition, and C3 adds the class descriptions, which gives the benchmark-direct prompt. Scores are on texts valid under all three prompts. The nine decision and commercial models form one family of 72 contrasts and CoPE a family of 8.

Table S6. Ablation macro-F1 and adjusted differences (points).

Model	Dataset	C1	C2	C3	C2−C1	Adjusted CI	C3−C2	Adjusted CI
Jev	Dynamic	.815	.835	.832	+2.0	[−0.0, +3.9]	−0.3	[−1.0, +0.3]
	MHS	.687	.704	.689	+1.7	[−0.5, +4.1]	−1.5	[−3.3, +0.1]
	HateXplain	.643	.686	.691	**+4.3**	[+2.4, +6.3]	+0.5	[−0.6, +1.7]
	HateCheck	.951	.973	.970	+2.1	[−0.4, +4.1]	−0.3	[−0.7, +0.0]
Laya	Dynamic	.680	.623	.639	**−5.8**	[−7.8, −3.8]	**+1.6**	[+0.4, +2.9]
	MHS	.677	.701	.692	**+2.3**	[+0.2, +4.4]	−0.8	[−2.4, +0.6]
	HateXplain	.718	.753	.753	**+3.5**	[+1.5, +5.6]	−0.0	[−1.3, +1.2]
	HateCheck	.744	.717	.709	**−2.8**	[−5.5, −0.1]	−0.8	[−2.1, +0.5]
Decider	Dynamic	.649	.610	.623	**−3.9**	[−5.9, −2.1]	**+1.3**	[+0.4, +2.3]
	MHS	.750	.812	.794	**+6.2**	[+3.9, +8.6]	**−1.8**	[−3.2, −0.4]
	HateXplain	.626	.596	.610	**−3.0**	[−4.9, −1.1]	+1.4	[−0.1, +2.9]
	HateCheck	.763	.769	.781	+0.7	[−2.5, +3.7]	+1.1	[−0.3, +2.6]
SemIf-2B	Dynamic	.503	.590	.584	**+8.8**	[+5.7, +11.9]	−0.6	[−1.7, +0.4]
	MHS	.664	.390	.442	**−27.3**	[−31.0, −23.3]	**+5.2**	[+3.7, +6.8]
	HateXplain	.586	.486	.548	**−10.0**	[−13.1, −6.9]	**+6.2**	[+4.0, +8.7]
	HateCheck	.619	.696	.656	+7.7	[−1.1, +16.7]	**−4.0**	[−6.8, −1.7]
SemIf-4B	Dynamic	.690	.687	.652	−0.2	[−2.9, +2.4]	**−3.6**	[−4.9, −2.2]
	MHS	.610	.723	.725	**+11.3**	[+8.6, +14.0]	+0.2	[−1.7, +2.0]
	HateXplain	.571	.645	.692	**+7.4**	[+4.8, +10.1]	**+4.7**	[+2.1, +7.3]
	HateCheck	.785	.799	.786	+1.5	[−2.1, +5.3]	−1.4	[−4.6, +1.8]
Bespoke-Nimble	Dynamic	.701	.775	.772	**+7.3**	[+4.8, +9.9]	−0.3	[−1.8, +1.3]
	MHS	.592	.604	.616	+1.2	[−0.6, +3.0]	+1.2	[−0.7, +3.3]
	HateXplain	.463	.463	.459	−0.0	[−2.2, +2.1]	−0.4	[−2.2, +1.3]
	HateCheck	.812	.818	.785	+0.6	[−3.3, +4.2]	−3.3	[−7.6, +0.4]
Luna Decisions	Dynamic	.807	.696	.716	**−11.1**	[−13.3, −9.1]	**+2.0**	[+0.4, +3.4]
	MHS	.724	.666	.658	**−5.8**	[−8.8, −2.9]	−0.7	[−2.8, +1.2]
	HateXplain	.790	.793	.776	+0.3	[−2.2, +3.0]	−1.6	[−4.4, +0.8]
	HateCheck	.948	.906	.911	**−4.2**	[−8.8, −1.1]	+0.5	[−0.5, +1.6]
CoPE	Dynamic	.834	.813	.810	**−2.1**	[−3.4, −0.8]	−0.2	[−0.9, +0.4]
	MHS	.672	.650	.653	**−2.2**	[−3.6, −0.8]	+0.3	[−0.8, +1.4]
	HateXplain	.606	.599	.587	−0.7	[−2.0, +0.6]	**−1.3**	[−2.3, −0.3]

Model	Dataset	C1	C2	C3	C2−C1	Adjusted CI	C3−C2	Adjusted CI
GPT-5.6 Luna	HateCheck	.934	.921	.913	**−1.3**	[−2.6, −0.2]	**−0.9**	[−1.7, −0.3]
	Dynamic	.857	.876	.873	**+1.9**	[+0.3, +3.6]	−0.4	[−1.5, +0.7]
	MHS	.719	.712	.695	−0.7	[−3.1, +1.7]	−1.7	[−3.6, +0.0]
	HateXplain	.680	.656	.670	**−2.4**	[−4.6, −0.4]	+1.4	[−0.5, +3.3]
GPT-6 Sol	HateCheck	.969	.978	.974	+0.8	[−1.4, +2.5]	−0.4	[−1.3, +0.7]
	Dynamic	.861	.898	.891	**+3.6**	[+2.1, +5.0]	−0.7	[−1.8, +0.3]
	MHS	.714	.699	.698	−1.5	[−3.5, +0.6]	−0.1	[−2.0, +1.7]
	HateXplain	.791	.783	.784	−0.8	[−2.9, +1.3]	+0.1	[−1.9, +2.0]
	HateCheck	.976	.979	.973	+0.3	[−0.6, +1.2]	−0.6	[−1.5, +0.1]

What this shows. In the main family, the definition (C2 − C1) gives 10 significant gains and 10 losses in 36 contrasts, and the class descriptions (C3 − C2) 6 gains and 3 losses. The largest effects follow the definition, such as −27.3 points for SemIf-2B and +11.3 for SemIf-4B on MHS, and −11.1 for Luna Decisions on Dynamic. The class descriptions mostly move macro-F1 by less than two points, and partly reverse the definition's effect for SemIf-2B on MHS and HateXplain. CoPE loses with the definition on three datasets, consistent with its losses in Section 4.2.

S7 Prompt Robustness

The paper reports the macro-F1 changes (Appendix F, Table 4). Table S7a adds the share of changed predictions and Table S7b their direction. Option-label variants rename the options to A and B (A/B) or reverse their order (Rev.), in a family of 10 contrasts. The paraphrases P1 and P2 replace the MHS definition, in a family of 20. All variants run on MHS under benchmark-direct classification.

Table S7a. Change in MHS macro-F1 (points) and share of changed predictions (%).

Model	Orig.	A/B Δ	A/B chg. %	Rev. Δ	Rev. chg. %	P1 Δ	P1 chg. %	P2 Δ	P2 chg. %
Laya	.692	**−25.2**	44.5	+0.7	5.5	−0.4	2.7	−0.4	1.9
Decider	.794	−0.1	3.5	+0.6	2.0	−0.8	3.0	**−1.1**	2.4
SemIf-2B	.442	**−1.9**	8.1	**−5.7**	6.9	**−1.3**	3.8	**−1.6**	2.8
SemIf-4B	.725	**−3.3**	11.1	+1.0	4.7	−1.2	4.8	+0.3	3.7
Bespoke-Nimble	.616	−0.5	9.2	**+5.2**	21.9	**−2.0**	7.9	−1.2	6.1
Jev	.689	–	–	–	–	+0.4	3.3	+0.6	2.5
Luna Decisions	.658	–	–	–	–	−1.0	3.8	−0.3	4.4
CoPE	.653	–	–	–	–	−0.1	2.2	−0.4	2.3
GPT-5.6 Luna	.695	–	–	–	–	+0.7	5.6	+0.8	5.2
GPT-6 Sol	.698	–	–	–	–	−1.4	4.2	−0.1	4.2

Table S7b. Changed predictions per variant, split into corrected and introduced false positives (FP) and false negatives (FN).

Model	Variant	Changed	Corr. FP	Corr. FN	Intr. FP	Intr. FN
Laya	A/B	1424	418	40	521	445
Laya	Rev.	177	68	31	54	24
Laya	P1	86	23	13	36	14
Laya	P2	60	15	9	24	12
Decider	A/B	112	57	4	19	32
Decider	Rev.	64	44	1	4	15
Decider	P1	97	24	13	41	19
Decider	P2	76	4	15	50	7
SemIf-2B	A/B	260	97	4	150	9
SemIf-2B	Rev.	222	18	9	195	0
SemIf-2B	P1	122	36	6	78	2
SemIf-2B	P2	89	17	3	68	1
SemIf-4B	A/B	354	188	0	7	159
SemIf-4B	Rev.	152	21	56	69	6
SemIf-4B	P1	153	54	11	49	39
SemIf-4B	P2	119	42	23	31	23
Bespoke-Nimble	A/B	296	159	7	39	91
Bespoke-Nimble	Rev.	702	519	0	1	182
Bespoke-Nimble	P1	253	75	20	116	42
Bespoke-Nimble	P2	194	31	35	122	6
Jev	P1	106	49	14	22	21
Jev	P2	80	33	16	20	11
Luna Decisions	P1	122	38	18	31	35
Luna Decisions	P2	141	31	34	42	34
CoPE	P1	71	22	9	37	3

Model	Variant	Changed	Corr. FP	Corr. FN	Intr. FP	Intr. FN
CoPE	P2	74	36	0	21	17
GPT-5.6 Luna	P1	180	90	17	42	31
GPT-5.6 Luna	P2	167	66	27	58	16
GPT-6 Sol	P1	133	57	7	19	50
GPT-6 Sol	P2	133	49	23	24	37

The paraphrases were frozen before inference, and only the definition paragraph differs from the benchmark-direct MHS prompt.

> **Original.** Hate speech is prejudice-driven, hostile or malicious language directed at a person or group because of actual or perceived innate characteristics, particularly when the group is unnecessarily labeled. Assess whether the comment contains hate speech under this definition; the available judgments are yes, no, and unclear.
>
> **P1 (reordered clauses).** Language is hate speech when it targets a person or group because of their actual or perceived innate characteristics and is driven by prejudice and hostile or malicious, especially when it labels the group without need. Decide whether the comment contains hate speech under this definition; the possible judgments are yes, no, and unclear.
>
> **P2 (synonyms).** Hate speech is language motivated by prejudice that is hostile or malicious toward a person or group on the basis of real or assumed innate traits, in particular where the group is labeled needlessly. Determine whether the comment includes hate speech according to this definition; the judgments available are yes, no, and unclear.

Model rankings correlate at Spearman ρ of .90 (original and P1), .98 (original and P2) and .95 (P1 and P2).

What this shows. Option names matter far more than definition wording. Renaming the options changes 44.5% of Laya's predictions, and reversing them 21.9% of Bespoke-Nimble's, while the paraphrases change at most 7.9% of any model's predictions. The paraphrases still change 1.9% to 7.9% of predictions, about as many as some dataset-specific definitions in Table S5, so part of each definition effect may reflect wording rather than content.

S8 Decomposition

S8.1 All Direct and Decomposed Comparisons

The paper summarizes 44 comparisons of direct (A) and decomposed (B) classification (Table 2, Figure 4). Table S8a lists them. Dynamic, MHS and HateXplain form the primary family of 30 contrasts, HateCheck a family of 10 and Luna Decisions a family of 4 on texts it answered under both conditions. ΔP and ΔR give the change in hate precision and recall in points. § marks the 3 comparisons in which A or B scores below labeling every text non-hate.

Table S8a. Decomposed minus direct macro-F1 on the same texts and definition.

Model	Dataset	N	A	B	B−A	Adjusted 95% CI	ΔP	ΔR
Jev	Dynamic	4,120	.826	.842	**+1.6**	[+0.5, +2.8]	−1.2	+4.2
Laya	Dynamic	4,120	.654	.533	**−12.1**	[−14.4, −9.8]	+4.6	−27.1
Decider	Dynamic	4,120	.585	.554	**−3.2**	[−5.3, −1.0]	+1.5	−6.0
SemIf-2B	Dynamic	4,120	.479	.536	**+5.7**	[+2.7, +8.7]	+1.2	−38.3
SemIf-4B	Dynamic	4,120	.675	.656	**−2.0**	[−3.6, −0.3]	+0.4	−4.7
Bespoke-Nimble	Dynamic	4,120	.742	.705	**−3.7**	[−5.5, −1.9]	−2.6	−3.5
Luna Decisions	Dynamic	4,098	.656	.762	**+10.6**	[+9.0, +12.1]	–	–
CoPE	Dynamic	4,120	.826	.825	−0.2	[−0.8, +0.4]	+0.6	−1.2
BART	Dynamic	4,120	.355	.393	**+3.8**	[+2.2, +5.3]	+0.2	−3.8
GPT-5.6 Luna	Dynamic	4,120	.858	.852	−0.6	[−2.4, +1.2]	−10.4	+12.3
GPT-6 Sol	Dynamic	4,120	.896	.895	−0.1	[−1.4, +1.3]	−5.1	+5.7
Jev	MHS	3,203	.689	.681	−0.8	[−2.0, +0.4]	−1.1	−1.0
Laya	MHS	3,203	.694	.733	**+3.9**	[+1.9, +5.9]	+6.0	−0.8
Decider	MHS	3,203	.800	.707	**−9.3**	[−12.3, −6.2]	+4.4	−37.0
SemIf-2B	MHS	3,203	.220	.579	**+35.9**§	[+33.3, +38.5]	+10.4	−25.4
SemIf-4B	MHS	3,203	.718	.683	**−3.6**	[−5.7, −1.5]	−2.6	−11.9
Bespoke-Nimble	MHS	3,203	.585	.551	**−3.4**	[−5.7, −1.1]	−4.4	−26.0
Luna Decisions	MHS	3,203	.608	.581	**−2.7**	[−4.5, −0.9]	–	–
CoPE	MHS	3,203	.676	.667	**−0.9**	[−1.7, −0.1]	−0.9	−1.5
BART	MHS	3,203	.196	.196	0.0§	[0.0, 0.0]	0.0	0.0
GPT-5.6 Luna	MHS	3,203	.692	.628	**−6.4**	[−8.5, −4.3]	−10.8	+5.6
GPT-6 Sol	MHS	3,203	.682	.682	+0.1	[−2.1, +2.3]	−5.6	+10.6
Jev	HateXplain	1,924	.721	.713	−0.8	[−2.0, +0.4]	−1.2	+1.3
Laya	HateXplain	1,924	.722	.768	**+4.6**	[+2.0, +7.4]	+8.4	−10.4
Decider	HateXplain	1,924	.656	.687	**+3.1**	[+0.2, +6.0]	+5.1	−6.7

Model	Dataset	N	A	B	B−A	Adjusted 95% CI	ΔP	ΔR
SemIf-2B	HateXplain	1,924	.418	.444	+2.6	[−0.3, +5.7]	−0.0	−6.6
SemIf-4B	HateXplain	1,924	.675	.677	+0.3	[−1.9, +2.4]	+0.8	−4.4
Bespoke-Nimble	HateXplain	1,924	.512	.529	+1.7	[−0.4, +3.6]	+0.5	−2.7
Luna Decisions	HateXplain	1,924	.775	.745	**−3.0**	[−4.8, −1.3]	–	–
CoPE	HateXplain	1,924	.634	.632	−0.3	[−1.0, +0.5]	−0.2	0.0
BART	HateXplain	1,924	.240	.242	+0.2§	[−0.8, +1.2]	+0.1	+0.2
GPT-5.6 Luna	HateXplain	1,924	.663	.611	**−5.2**	[−7.6, −3.0]	−3.9	+2.0
GPT-6 Sol	HateXplain	1,924	.768	.719	**−4.9**	[−7.3, −2.4]	−9.2	+9.4
Jev	HateCheck	3,728	.960	.957	−0.4	[−1.5, +0.7]	−0.7	+0.3
Laya	HateCheck	3,728	.701	.690	−1.1	[−4.3, +2.2]	+3.0	−7.7
Decider	HateCheck	3,728	.730	.390	**−34.0**	[−38.4, −29.6]	−0.8	−59.3
SemIf-2B	HateCheck	3,728	.483	.411	−7.2	[−15.0, +1.0]	+5.1	−76.7
SemIf-4B	HateCheck	3,728	.782	.754	**−2.8**	[−5.0, −0.7]	−0.1	−5.2
Bespoke-Nimble	HateCheck	3,728	.780	.785	+0.5	[−1.0, +2.1]	+0.4	+0.2
Luna Decisions	HateCheck	3,719	.904	.899	−0.5	[−1.6, +0.6]	–	–
CoPE	HateCheck	3,728	.936	.940	+0.4	[−0.1, +1.0]	+0.6	−0.2
BART	HateCheck	3,728	.407	.407	0.0	[0.0, 0.0]	0.0	0.0
GPT-5.6 Luna	HateCheck	3,728	.974	.950	**−2.4**	[−5.0, −0.0]	−3.5	+0.7
GPT-6 Sol	HateCheck	3,728	.976	.978	+0.2	[−0.4, +0.8]	+0.0	+0.2

In the primary family, decomposition gives 7 significant gains, 11 losses and 12 inconclusive results. Without the 3 marked comparisons, it gives 6 gains and 11 losses. On HateCheck it gives no gain and 3 losses, including −34.0 points for Decider, whose recall falls by 59 points.

S8.2 Threshold Control and Soft Aggregation

The paper's Table 5 reports A, B, A* and B* with the B − A* and B* − A* contrasts. Table S8b adds the soft rule and the effect of tuning itself. The primary family (P) holds Laya, SemIf-4B, Jev and Luna Decisions, and the exploratory family (E) the other four models, with adjustment within families of 24 contrasts. § marks the two Luna Decisions cells fitted post hoc on the 199 development texts with complete outputs.

Table S8b. Macro-F1 under each operating point, and adjusted differences (points).

Model	Dataset	Fam.	A	B	A*	B*	Soft	A*−A	B*−B	Soft−A*	Soft−B*
Laya	Dynamic	P	.654	.533	.649	.654	.658	−0.4	**+12.1**	+0.9	+0.4
	MHS	P	.694	.733	.695	.718	.727	+0.1	−1.4	**+3.2**	+0.9
	HateXplain	P	.722	.768	.781	.782	.793	**+5.9**	+1.4	+1.2	+1.1
SemIf-4B	Dynamic	P	.675	.656	.719	.705	.713	**+4.3**	**+5.0**	−0.6	+0.7
	MHS	P	.718	.683	.710	.718	.702	−0.9	**+3.5**	−0.7	**−1.6**
	HateXplain	P	.675	.677	.724	.697	.692	**+5.0**	+2.0	**−3.3**	−0.5
Jev	Dynamic	P	.826	.842	.852	.821	.826	**+2.6**	**−2.1**	**−2.7**	+0.5
	MHS	P	.689	.681	.690	.671	.683	+0.1	−1.0	−0.7	**+1.2**
	HateXplain	P	.721	.713	.755	.720	.710	**+3.5**	+0.7	**−4.6**	−1.0
Luna Decisions	Dynamic§	P	.656	.763	.774	.768	.810	**+11.8**	+0.6	**+3.6**	**+4.1**
	MHS	P	.608	.581	.623	.629	.609	**+1.6**	**+4.8**	−1.4	**−2.0**
	HateXplain§	P	.775	.745	.770	.753	.744	−0.4	+0.9	−2.6	−0.9
Decider	Dynamic	E	.585	.554	.689	.692	.671	**+10.3**	**+13.9**	−1.7	**−2.1**
	MHS	E	.800	.707	.805	.763	.741	+0.5	**+5.5**	**−6.4**	−2.2
	HateXplain	E	.656	.687	.695	.704	.705	**+3.9**	+1.7	+1.0	+0.1
SemIf-2B	Dynamic	E	.479	.536	.549	.535	.527	**+7.0**	−0.0	**−2.2**	−0.9
	MHS	E	.220	.579	.620	.649	.610	**+40.1**	**+7.0**	−1.0	**−3.9**
	HateXplain	E	.418	.444	.608	.594	.547	**+19.0**	**+15.1**	**−6.1**	**−4.8**
Bespoke-Nimble	Dynamic	E	.742	.705	.765	.730	.730	**+2.3**	**+2.5**	**−3.5**	+0.1
	MHS	E	.585	.551	.619	.681	.591	+3.5	**+13.1**	**−2.8**	**−9.0**
	HateXplain	E	.512	.529	.729	.700	.688	**+21.6**	**+17.1**	**−4.0**	−1.1
BART	Dynamic	E	.355	.393	.472	.506	.485	**+11.7**	**+11.4**	+1.3	**−2.1**
	MHS	E	.196	.196	.522	.536	.547	**+32.6**	**+33.9**	+2.5	+1.1
	HateXplain	E	.240	.242	.487	.496	.469	**+24.7**	**+25.5**	−1.8	−2.7

Table S8c. Selected thresholds and validation macro-F1. B* tunes one threshold per binary component, and Dynamic's conduct answer keeps the highest score.

Model	Dataset	A* threshold	B* thresholds	Soft threshold	Val. A*	Val. B*	Val. soft
Laya	Dynamic	0.522	scope 0.351, intentional 0.112	0.080	.719	.679	.665
	MHS	0.808	targeting 0.827, abuse 0.531	0.617	.733	.808	.795
	HateXplain	0.821	targeting 0.329, abuse 0.764	0.638	.823	.816	.801
SemIf-4B	Dynamic	0.223	scope 0.378, intentional 0.500	0.190	.710	.703	.710
	MHS	0.731	targeting 0.294, abuse 0.755	0.369	.774	.800	.763

Model	Dataset	A* threshold	B* thresholds	Soft threshold	Val. A*	Val. B*	Val. soft
	HateXplain	0.893	targeting 0.679, abuse 0.881	0.644	.738	.749	.735
Jev	Dynamic	0.320	scope 0.640, intentional 0.660	0.403	.845	.859	.845
	MHS	0.700	targeting 0.660, abuse 0.500	0.453	.771	.786	.776
	HateXplain	0.830	targeting 0.900, abuse 0.890	0.881	.779	.779	.755
Luna Decisions	Dynamic§	0.110	scope 0.500, intentional 0.140	0.056	.796	.786	.814
	MHS	0.430	targeting 0.070, abuse 0.600	0.144	.734	.698	.680
	HateXplain§	0.620	targeting 0.280, abuse 0.530	0.145	.804	.775	.771
Decider	Dynamic	0.177	scope 0.184, intentional 0.306	0.152	.690	.690	.684
	MHS	0.629	targeting 0.099, abuse 0.861	0.210	.882	.855	.805
	HateXplain	0.751	targeting 0.572, abuse 0.651	0.467	.698	.725	.713
SemIf-2B	Dynamic	0.881	scope 0.469, intentional 0.622	0.284	.625	.585	.564
	MHS	0.947	targeting 0.469, abuse 0.867	0.444	.699	.721	.679
	HateXplain	0.905	targeting 0.593, abuse 0.777	0.750	.636	.615	.604
Bespoke-Nimble	Dynamic	0.881	scope 0.500, intentional 0.881	0.610	.739	.749	.729
	MHS	0.971	targeting 0.023, abuse 0.940	0.933	.681	.745	.658
	HateXplain	0.971	targeting 0.905, abuse 0.947	0.912	.757	.746	.750
BART	Dynamic	0.832	scope 0.750, intentional 0.500	0.509	.543	.540	.516
	MHS	0.849	targeting 0.773, abuse 0.758	0.596	.574	.605	.583
	HateXplain	0.718	targeting 0.807, abuse 0.746	0.660	.551	.570	.563

Tuning the direct threshold (A* − A) gives 17 significant gains in 24 comparisons, and tuning the components (B* − B) 14 gains. The soft rule beats A* in 2 comparisons and loses in 9, and beats B* in 2 and loses in 7.

S8.3 Do Corrected Verdicts Rest on Correct Components?

Among the 200 audited texts (Section S11), decomposition corrects a direct verdict in 140 model-text pairs. Table S8d checks each corrected pair against the audit reference's component answers.

Table S8d. Audited model-text pairs that decomposition corrects, by whether every component answer matches the audit reference. Unresolved pairs have a component without a reference answer.

Model	Corrected pairs	All components correct	Some component wrong	Unresolved	FP corrected	FN corrected
Jev	2	2	0	0	2	0
Laya	19	6	12	1	15	4
Decider	21	7	13	1	19	2
SemIf-2B	43	13	28	2	43	0
SemIf-4B	19	11	7	1	15	4
Bespoke-Nimble	14	6	6	2	12	2
Luna Decisions	8	4	4	0	0	8
CoPE	3	2	1	0	0	3
BART	1	0	1	0	1	0
GPT-5.6 Luna	7	6	1	0	2	5
GPT-6 Sol	3	3	0	0	0	3
Total	140	60	73	7	109	31

S8.4 Luna Decisions Refusals

Luna Decisions refuses some requests, mostly on Dynamic and HateCheck (Table S8e). The paper excludes refusals. Table S8f recomputes the decomposition contrasts with refusals counted as hate or as errors, that is, as the label opposite to the gold label.

Table S8e. Luna Decisions requests and refusals by condition. Across all test conditions, including attributes and the three-class task, it refused 135 of 85,585 requests.

Dataset	Condition	Requested	Refusals
Dynamic	Bench.	4,120	9
Dynamic	No def.	4,120	0
Dynamic	A	4,120	22
Dynamic	B	4,120	2
MHS	Bench.	3,203	0
MHS	No def.	3,203	0
MHS	A	3,203	0
MHS	B	3,203	0
HateXplain	Bench.	1,924	0
HateXplain	No def.	1,924	0
HateXplain	A	1,924	0
HateXplain	B	1,924	0
HateCheck	Bench.	3,728	9

Dataset	Condition	Requested	Refusals
HateCheck	No def.	3,728	0
HateCheck	A	3,728	9
HateCheck	B	3,728	0

Table S8f. Luna Decisions decomposition contrasts under three treatments of refusals (adjusted within the family of 4).

Dataset	Refusals	A	B	B−A	Adjusted 95% CI
Dynamic	excluded	.656	.762	**+10.6**	[+9.0, +12.1]
Dynamic	as hate	.657	.762	**+10.5**	[+9.0, +12.1]
Dynamic	as error	.652	.762	**+11.0**	[+9.5, +12.6]
MHS	excluded	.608	.581	**−2.7**	[−4.5, −0.9]
MHS	as hate	.608	.581	**−2.7**	[−4.5, −0.9]
MHS	as error	.608	.581	**−2.7**	[−4.5, −0.9]
HateXplain	excluded	.775	.745	**−3.0**	[−4.8, −1.3]
HateXplain	as hate	.775	.745	**−3.0**	[−4.8, −1.3]
HateXplain	as error	.775	.745	**−3.0**	[−4.8, −1.3]
HateCheck	excluded	.904	.899	−0.5	[−1.6, +0.6]
HateCheck	as hate	.904	.898	−0.6	[−1.7, +0.5]
HateCheck	as error	.901	.898	−0.3	[−1.4, +0.8]

What this shows. Decomposition is not a reliable improvement. Most of its gains come from cells where the direct prompt was badly calibrated, and a threshold tuned on 200 texts recovers them (A* − A gains in 17 of 24 comparisons). Combining component scores softly does not help either, so the hard conjunctive rule is not what holds decomposition back. Of the 133 corrected pairs with resolved components, 73 rest on at least one wrong component, so decomposition often reaches the right verdict for the wrong reason. Treating refusals as hate or as errors changes no Luna Decisions result by more than 0.4 points, and no significance decision.

S9 Efficiency

Figure 3 of the paper shows five models. Table S9a and Figure S3 cover every model on the same 3,728 HateCheck texts. Latencies are median and 95th-percentile complete-decision times in milliseconds, with one request at a time. Hosted cost is the recorded charge per 1K texts. * marks local estimates, which price the measured serial throughput at $0.20 per GPU-hour.

Table S9a. Quality, latency and cost of direct (A) and decomposed (B) classification on HateCheck.

Model	F1 A	F1 B	A med.	A p95	B med.	B p95	Δ med.	Texts/s A	$/1K A	$/1K B	Δ cost
Laya	.701	.690	48	93	52	94	+9%	18.7	0.003*	0.003*	+7%
BART	.407	.407	35	74	47	66	+34%	24.1	0.002*	0.003*	+20%
SemIf-2B	.483	.411	125	272	251	540	+100%	6.6	0.008*	0.017*	+100%
Decider	.730	.390	134	293	135	265	+1%	6.3	0.009*	0.009*	−2%
SemIf-4B	.782	.754	184	325	344	556	+87%	4.8	0.012*	0.021*	+84%
CoPE	.936	.940	266	277	533	549	+100%	3.8	0.015*	0.030*	+100%
Bespoke-Nimble	.780	.785	335	642	686	1,255	+105%	2.6	0.021*	0.043*	+101%
Luna Decisions	.904	.898	243	573	291	655	+20%	3.4	0.057	0.113	+99%
Jev	.960	.957	336	537	343	622	+2%	2.6	0.033	0.054	+64%
GPT-5.6 Luna	.974	.950	844	1,273	885	1,303	+5%	1.1	0.120	0.129	+7%
GPT-6 Sol	.976	.978	1,426	2,124	1,383	2,294	−3%	0.7	1.177	1.256	+7%

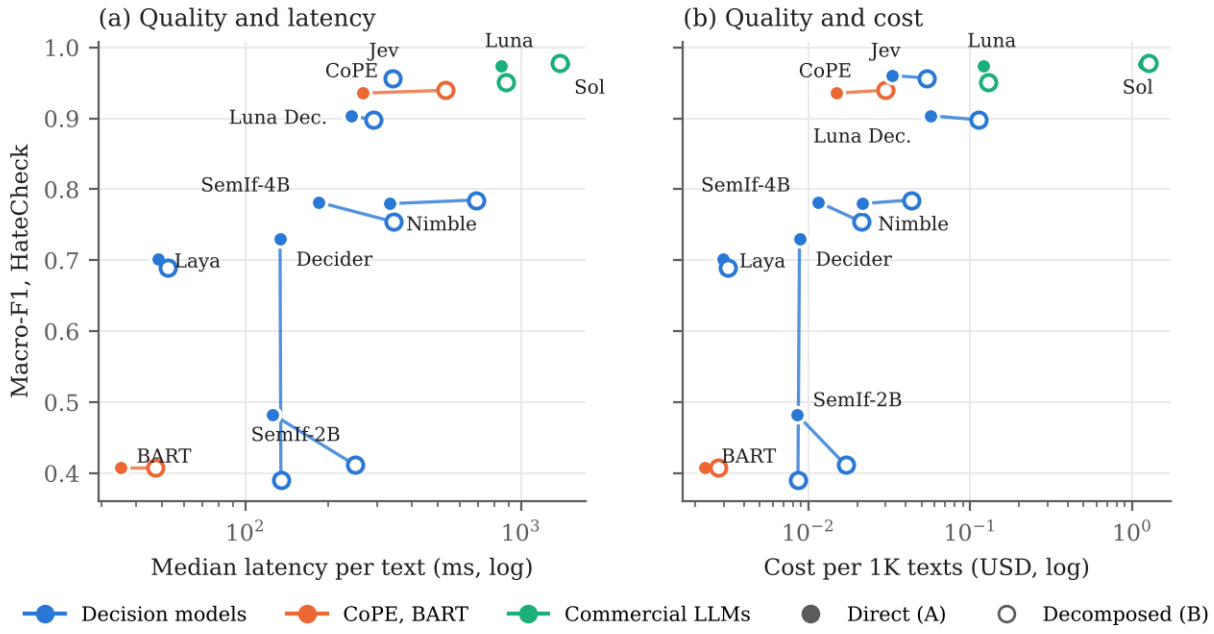


Figure S3. Macro-F1 against median latency (a) and cost per 1K texts (b) on HateCheck for every model. Filled markers show direct and hollow markers decomposed classification. Local costs are estimates at $0.20 per GPU-hour.

Table S9b. Latency and cost on the other datasets. Jev, Laya, SemIf-4B and Sol were timed on a common sample of 800 texts per dataset, and the other models on the full test sets. Luna Decisions was timed with four concurrent requests and its cost is the recorded charge on the full test set. Luna was timed on only 100 texts per dataset and is omitted.

Dataset	Model	Texts	A med.	B med.	Δ med.	$/1K A	$/1K B	Δ cost
Dynamic	Laya	800	49	54	+12%	0.003*	0.003*	+12%
	BART	full	36	117	+224%	0.002*	0.007*	+171%
	SemIf-2B	full	142	447	+214%	0.009*	0.028*	+207%
	Decider	full	148	160	+8%	0.010*	0.010*	−1%
	SemIf-4B	800	183	495	+171%	0.012*	0.030*	+157%
	CoPE	full	289	2,320	+703%	0.016*	0.130*	+704%
	Bespoke-Nimble	full	367	1,154	+215%	0.023*	0.069*	+203%
	Luna Decisions	full	254	255	+0%	0.070	0.206	+196%
	Jev	800	304	308	+1%	0.039	0.093	+139%
	GPT-6 Sol	800	1,587	1,825	+15%	1.438	1.616	+12%
MHS	Laya	800	39	41	+6%	0.003*	0.003*	+3%
	BART	full	33	36	+6%	0.002*	0.002*	+4%
	SemIf-2B	full	108	215	+100%	0.007*	0.015*	+100%
	Decider	full	115	114	−1%	0.008*	0.008*	−3%
	SemIf-4B	800	133	242	+82%	0.009*	0.017*	+79%

Dataset	Model	Texts	A med.	B med.	Δ med.	$/1K A	$/1K B	Δ cost
	CoPE	full	211	424	+101%	0.013*	0.026*	+100%
	Bespoke-Nimble	full	294	594	+102%	0.019*	0.038*	+100%
	Luna Decisions	full	245	272	+11%	0.038	0.072	+89%
	Jev	800	308	307	−0%	0.025	0.036	+47%
	GPT-6 Sol	800	1,712	1,673	−2%	0.802	0.874	+9%
HateXplain	Laya	800	39	42	+9%	0.003*	0.003*	+6%
	BART	full	34	38	+13%	0.002*	0.002*	+9%
	SemIf-2B	full	116	232	+100%	0.008*	0.016*	+101%
	Decider	full	122	124	+2%	0.008*	0.008*	−3%
	SemIf-4B	800	143	283	+97%	0.010*	0.018*	+88%
	CoPE	full	223	454	+104%	0.014*	0.028*	+101%
	Bespoke-Nimble	full	313	635	+103%	0.020*	0.040*	+101%
	Luna Decisions	full	249	315	+27%	0.046	0.089	+95%
	Jev	800	305	304	−0%	0.028	0.043	+56%
	GPT-6 Sol	800	1,682	1,666	−1%	0.952	1.028	+8%

Local models ran on one NVIDIA T4 GPU (16 GB), with batch size 1 and one request at a time. The local estimates therefore price the measured T4 time at the highest listed rental rate, and batching would lower them. Hosted latencies include network round trips from our server, so they depend on its location.

What this shows. Decomposition roughly doubles latency and local cost for SemIf, Bespoke-Nimble and CoPE. For Laya, Decider, Jev, Luna and Sol, median latency changes by at most 9% on HateCheck, but hosted cost still rises because each component is billed, by 47% to 139% for Jev and 89% to 196% for Luna Decisions. On Dynamic, CoPE answers the conduct question as six judgments, which raises its latency eightfold. Under direct classification every local model costs less per text than Jev, but only Jev, Luna Decisions and CoPE exceed .90 on HateCheck at under $0.06 per 1K texts.

S10 Cascades

Each cascade refers the first model's least confident texts to a second model, at a referral rate frozen on validation texts (Appendix I). Table S10a lists all 28 frozen points with their comparisons, from one family of 84 contrasts. TOST gives the 90% interval of the difference from the second model, and the last column of that cell states whether the interval lies within one point ($\leq$1) or two points ($\leq$2) of zero.

Table S10a. Frozen cascade operating points under benchmark-direct classification.

Cascade	Dataset	Referred	Macro-F1	vs first	vs second	vs random	TOST 90% CI vs second	Hosted $/1K
Laya→Jev	Dynamic	100%	.832	**+19.3**	0.0	+0.0	[0.00, 0.00] $\leq$1	0.022
	MHS	5%	.698	+0.6	+0.9	+0.6	[−0.58, +2.46] no	0.001
	HateXplain	0%	.753	0.0	**+6.2**	+0.0	[+4.41, +7.98] no	0.000
	HateCheck	100%	.970	**+26.1**	0.0	0.0	[0.00, 0.00] $\leq$1	0.022
Laya→Luna Dec.	Dynamic	100%	.716	**+7.7**	0.0	−0.0	[0.00, 0.00] $\leq$1	0.032
	MHS	100%	.658	−3.4	0.0	0.0	[0.00, 0.00] $\leq$1	0.025
	HateXplain	20%	.779	**+2.6**	+0.3	+2.1	[−1.35, +1.92] $\leq$2	0.006
	HateCheck	100%	.909	**+20.0**	0.0	+0.0	[0.00, 0.00] $\leq$1	0.032
Laya→Luna	Dynamic	100%	.873	**+23.4**	0.0	−0.0	[0.00, 0.00] $\leq$1	0.068
	MHS	100%	.695	+0.3	0.0	+0.0	[0.00, 0.00] $\leq$1	0.054
	HateXplain	0%	.753	0.0	**+8.3**	+0.0	[+6.49, +10.16] no	0.000
	HateCheck	100%	.974	**+26.5**	0.0	+0.0	[0.00, 0.00] $\leq$1	0.068
Laya→Sol	Dynamic	100%	.891	**+25.2**	0.0	−0.0	[0.00, 0.00] $\leq$1	0.654
	MHS	100%	.698	+0.5	0.0	0.0	[0.00, 0.00] $\leq$1	0.511
	HateXplain	20%	.777	**+2.4**	−0.7	+1.8	[−2.28, +0.80] no	0.127
	HateCheck	100%	.973	**+26.4**	0.0	+0.0	[0.00, 0.00] $\leq$1	0.652
Jev→Luna Dec.	Dynamic	5%	.828	−0.4	**+11.1**	+0.2	[+10.14, +12.17] no	0.024
	MHS	20%	.696	+0.7	**+3.8**	+1.1	[+2.52, +5.01] no	0.024
	HateXplain	100%	.776	**+8.5**	0.0	+0.0	[0.00, 0.00] $\leq$1	0.053
	HateCheck	5%	.957	−1.2	**+4.9**	−0.9	[+3.42, +6.44] no	0.024
Jev→Luna	Dynamic	5%	.843	**+1.2**	**−3.0**	**+0.9**	[−3.70, −2.19] no	0.026
	MHS	20%	.702	+1.3	+0.7	+1.2	[−0.14, +1.54] $\leq$2	0.030
	HateXplain	20%	.688	−0.3	+1.8	+0.1	[+0.90, +2.72] no	0.035
	HateCheck	20%	.974	+0.4	−0.0	+0.3	[−0.14, +0.09] $\leq$1	0.036
Jev→Sol	Dynamic	50%	.888	**+5.6**	−0.3	**+2.7**	[−0.53, −0.04] $\leq$1	0.349
	MHS	20%	.701	+1.2	+0.4	+1.0	[−0.53, +1.25] $\leq$2	0.121
	HateXplain	100%	.784	**+9.3**	0.0	−0.0	[0.00, 0.00] $\leq$1	0.656
	HateCheck	50%	.973	+0.3	0.0	+0.1	[−0.07, +0.06] $\leq$1	0.348

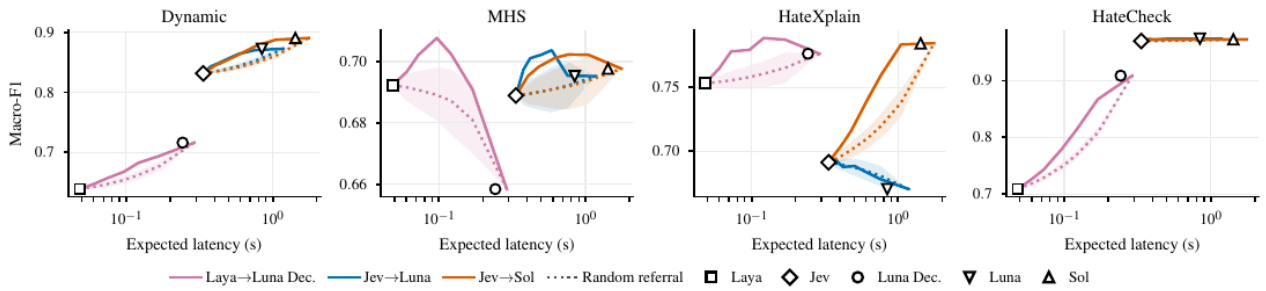


Figure S4. Macro-F1 against expected latency at every referral rate. Dotted lines show random referral.

Table S10b. Estimated HateCheck latency at the frozen operating points. Each text adds the first model's measured time and, if referred, the second model's.

Cascade	Referred texts	Median s	Mean s	95th pct. s	Hosted $ / 1K texts
Jev→Sol	1,864 / 3,728	1.467	1.118	2.194	0.348
Jev→Luna	746 / 3,728	0.335	0.558	1.451	0.036
Jev→Luna Dec.	186 / 3,728	0.327	0.356	0.550	0.024
Laya→Jev	3,728 / 3,728	0.371	0.388	0.461	0.022
Laya→Sol	3,728 / 3,728	1.525	1.608	2.157	0.652
Laya→Luna	3,728 / 3,728	1.021	1.098	1.523	0.068
Laya→Luna Dec.	3,728 / 3,728	0.297	0.333	0.489	0.032

To check the estimate, we ran Jev→Sol live on all HateCheck texts, one request at a time. It reached .976 macro-F1 at the same hosted cost ($0.348 per 1K texts), with a median service time of 1.022 s against the estimated 1.467 s. The live run referred 1,789 of the 1,864 texts referred offline, and its labels agree with the cached labels on 99.7% of texts for Jev and 98.6% for Sol.

Table S10c. Exploratory routing from Laya to Sol under direct classification (A), on an earlier common sample of 800 texts per dataset. Corrected and introduced count the referred texts whose label Sol corrects or breaks.

Dataset	Referred	Macro-F1	P	R	Corrected	Introduced	Hosted $ (800 texts)
Dynamic	0%	.658	.748	.573	0	0	0.000
	5%	.671	.763	.584	12	2	0.057
	10%	.690	.781	.607	30	5	0.114
	20%	.729	.812	.659	62	6	0.229
	30%	.772	.852	.709	100	9	0.344
	50%	.819	.891	.764	143	15	0.574
	100%	.888	.938	.855	216	32	1.150
MHS	0%	.696	.481	.718	0	0	0.000
	5%	.701	.493	.697	13	6	0.032
	10%	.712	.513	.687	27	9	0.064
	20%	.721	.527	.692	47	22	0.128
	30%	.726	.543	.677	64	32	0.192
	50%	.730	.580	.610	101	57	0.321
	100%	.689	.567	.477	129	96	0.642
HateXplain	0%	.730	.547	.899	0	0	0.000
	5%	.740	.559	.895	12	3	0.038
	10%	.753	.576	.895	27	7	0.077
	20%	.763	.590	.891	45	16	0.152
	30%	.779	.610	.899	67	25	0.229
	50%	.780	.616	.879	91	47	0.381
	100%	.770	.634	.777	126	83	0.762

What this shows. Of the 28 frozen points, 15 refer all or no texts, where cascade and random referral coincide, and confidence-based referral is significantly better than random only twice. No cascade significantly exceeds its stronger model, so cascades buy cost, not quality. Jev→Sol on Dynamic and Laya→Luna Decisions on HateXplain are the useful cases, since both are equivalent to their stronger model within the stated margin at half or a fifth of its hosted cost. Routing more texts can also hurt. On MHS, sending every text from Laya to Sol lowers macro-F1 from .730 at 50% to .689, because Sol corrects 129 labels but breaks 96.

S11 Policy and Component Audit

The audit reference for each judgment is the agreement of at least two of the expert and the two LLM judges (Appendix J). Table S11a gives agreement between raters on mutually definite answers, and Table S11b the model agreement behind Table 3 of the paper, under the majority and under the expert alone. Its intervals come from an unstratified bootstrap over audit texts that keeps the inclusion weights fixed.

Table S11a. Rater agreement on mutually definite answers, as agreeing/compared (Cohen κ). Component κ is not pooled across questions.

Dataset	Judgment	Expert–Opus	Expert–Gemini	Opus–Gemini
All	policy	174/182 (κ 0.91)	170/180 (κ 0.89)	175/183 (κ 0.91)
All	all components	371/410	364/423	356/412
Dynamic	policy	45/47 (κ 0.91)	46/47 (κ 0.96)	46/50 (κ 0.84)
	scope	41/48 (κ 0.64)	41/48 (κ 0.68)	42/50 (κ 0.65)
	intentional	44/45 (κ 0.94)	46/49 (κ 0.86)	42/46 (κ 0.80)
	conduct	40/49 (κ 0.73)	39/49 (κ 0.69)	40/50 (κ 0.70)
MHS	policy	40/42 (κ 0.90)	35/39 (κ 0.75)	37/39 (κ 0.87)
	targeting	40/42 (κ 0.90)	39/43 (κ 0.79)	36/39 (κ 0.83)
	abuse	41/42 (κ 0.94)	25/42 (κ 0.30)	25/43 (κ 0.29)
HateXplain	policy	42/45 (κ 0.86)	42/46 (κ 0.82)	46/46 (κ 1.00)
	targeting	39/43 (κ 0.78)	41/47 (κ 0.71)	41/42 (κ 0.95)
	abuse	40/44 (κ 0.80)	43/47 (κ 0.83)	40/45 (κ 0.77)
HateCheck	policy	47/48 (κ 0.96)	47/48 (κ 0.96)	46/48 (κ 0.92)
	targeting	40/48 (κ 0.65)	45/49 (κ 0.83)	42/48 (κ 0.75)
	abuse	46/49 (κ 0.88)	45/49 (κ 0.84)	48/49 (κ 0.96)

Table S11b. Model agreement with the audit reference (%, 95% interval). Agree uses the 187 resolved policy judgments, raw and reweighted by inverse inclusion probability, Conflicts the texts where the reference and the dataset label disagree (29 under the majority, 34 under the expert), and Comp. the texts whose component answers all resolve (180 and 187).

Model	Agree (maj.)	Reweighted (maj.)	Conflicts (maj.)	Comp. (maj.)	Agree (expert)	Reweighted (expert)	Conflicts (expert)	Comp. (expert)
Jev	89 (84–93)	93 (89–96)	72 (55–88)	72 (65–78)	90 (85–94)	93 (89–96)	74 (58–88)	70 (63–77)
Laya	61 (54–68)	65 (56–73)	45 (27–63)	39 (32–46)	61 (54–68)	63 (54–72)	41 (25–58)	37 (30–44)
Decider	64 (57–71)	73 (65–80)	31 (14–48)	31 (24–38)	64 (57–71)	72 (64–79)	32 (17–49)	30 (24–37)
SemIf-2B	48 (41–55)	58 (49–67)	28 (12–45)	27 (20–33)	49 (42–56)	59 (50–67)	29 (14–45)	27 (21–34)
SemIf-4B	71 (64–77)	77 (70–84)	55 (37–73)	52 (45–60)	69 (62–75)	75 (68–82)	47 (30–64)	51 (44–58)
Bespoke-Nimble	65 (58–72)	73 (65–80)	48 (30–67)	51 (43–58)	67 (60–74)	75 (68–81)	53 (36–70)	51 (44–58)
Luna Decisions	81 (75–86)	84 (77–89)	66 (47–83)	57 (50–65)	78 (72–84)	81 (74–87)	56 (39–72)	54 (47–61)
CoPE	78 (72–84)	84 (78–89)	62 (43–79)	57 (50–64)	80 (74–85)	84 (79–89)	68 (52–83)	56 (48–63)
BART	53 (46–61)	64 (55–71)	31 (14–48)	36 (29–43)	52 (45–60)	62 (53–70)	26 (12–42)	37 (30–44)
GPT-5.6 Luna	89 (85–94)	92 (87–95)	76 (59–91)	63 (56–70)	90 (85–94)	92 (88–96)	76 (61–90)	67 (60–74)
GPT-6 Sol	93 (89–96)	96 (93–98)	72 (55–88)	76 (70–82)	92 (88–96)	95 (92–98)	71 (54–86)	75 (69–81)

Table S11c. Texts on which the reference conflicts with the dataset label (%, 95% interval), raw and reweighted by inverse inclusion probability.

Dataset	Majority, raw	Majority, reweighted	Expert, raw	Expert, reweighted
Dynamic	2/49 = 4 (0–11)	3 (0–9)	1/47 = 2 (0–7)	2 (0–7)
MHS	17/44 = 39 (24–53)	21 (11–35)	18/45 = 40 (26–55)	24 (13–40)
HateXplain	9/46 = 20 (9–32)	17 (6–29)	13/46 = 28 (16–42)	28 (15–44)
HateCheck	1/48 = 2 (0–7)	2 (0–5)	2/49 = 4 (0–10)	3 (0–8)

Table S11c uses the same unstratified bootstrap for both references, so their intervals are comparable. The paper's main-text majority intervals (12% to 32% for MHS, 7% to 28% for HateXplain) come from the original analysis, which poststratifies each bootstrap draw to the sampling strata. Both LLM judges overrule a definite expert judgment on only 4 of 187 texts, and 22 of the 29 majority conflicts hold where the expert and both judges agree.

Table S11d. Dynamic texts selected as disputed examples, with each rater's policy judgment. The dataset texts are not reproduced.

Item	Dataset label	Expert	Opus	Gemini	Reference	Reference components
Dynamic:acl19236	hate	hate	hate	hate	hate	scope=yes; intentional=yes; conduct=dehumanization
Dynamic:acl19276	hate	hate	hate	hate	hate	scope=yes; intentional=yes; conduct=threatening
Dynamic:acl29276	hate	non-hate	hate	non-hate	non-hate	scope=no; intentional=yes; conduct=none

Item	Dataset label	Expert	Opus	Gemini	Reference	Reference components
Dynamic:acl29918	hate	hate	hate	non-hate	hate	scope=yes; intentional=yes; conduct=animosity
Dynamic:acl9874	hate	hate	hate	hate	hate	scope=yes; intentional=yes; conduct=animosity
Dynamic:acl19644	non-hate	uncertain	non-hate	non-hate	non-hate	scope=no; intentional=yes; conduct=derogation
Dynamic:acl19669	non-hate	non-hate	non-hate	non-hate	non-hate	scope=yes; intentional=no; conduct=derogation
Dynamic:acl19757	non-hate	non-hate	non-hate	non-hate	non-hate	scope=no; intentional=yes; conduct=none
Dynamic:acl29178	non-hate	uncertain	hate	non-hate	unresolved	scope=yes; intentional=unresolved; conduct=unresolved
Dynamic:acl29490	non-hate	non-hate	non-hate	non-hate	non-hate	scope=no; intentional=yes; conduct=none

What this shows. The expert and both LLM judges agree on 94% to 96% of mutually definite policy judgments (κ .89 to .91). Disagreement concentrates in components, above all MHS abuse, where Gemini disagrees with both the expert and Opus (κ .30 and .29). Taking the majority therefore protects the reference against one judge's reading. Model agreement barely depends on the reference, as the expert-only and majority values differ by at most 10 points on the conflict texts and 4 points elsewhere, and the ranking is nearly the same (Spearman ρ .99). The expert alone finds similar or higher conflict rates on MHS and HateXplain (24% and 28% against 21% and 17%), so the label conflicts do not come from the LLM judges.

S12 Policy Contrast

The policy contrast tests whether a model follows a controlled change of policy. P_{in} is HateCheck's definition, and P_{out} adds Dynamic's exclusion of men, white people and heterosexuals as hate targets. Each of the 90 switch texts should be hate under P_{in} and non-hate under P_{out}, and the 150 controls should keep their label. The constructed texts are in the repository, behind a content warning.

Table S12a. Switch and control rates on all 240 texts (%), and the switch rate on the 234 texts accepted by the annotation rule (Appendix K). Intervals are bootstrapped over templates. False flips are controls whose label changes.

Model	Switch % (95% CI)	Hateful controls %	Neutral controls %	Hateful false flips %	Neutral false flips %	Switch %, 234 accepted	Δ (pp)
Jev	90 (84–95)	100	100	0	0	96	+6.4
Laya	0 (0–0)	71	100	7	0	0	0.0
Decider	24 (18–31)	98	100	2	0	24	−0.6
SemIf-2B	12 (5–20)	98	88	2	10	11	−1.5
SemIf-4B	64 (53–76)	91	100	0	0	65	+1.0
Bespoke-Nimble	47 (40–55)	98	98	1	2	50	+3.3
Luna Decisions	71 (63–79)	97	100	1	0	76	+5.1
CoPE	58 (42–73)	100	70	0	7	61	+2.9
BART	0 (0–0)	100	97	0	0	0	0.0
GPT-5.6 Luna	83 (76–90)	97	100	2	0	89	+6.0
GPT-6 Sol	98 (94–100)	100	100	0	0	100	+2.2

Table S12b. Switch texts by targeted identity (30 each), as P_{in} accuracy / switch rate among texts labeled hate under P_{in} / switch rate among all texts (%).

Model	Men	White people	Straight people	All 90
Jev	70 / 100 / 70	100 / 100 / 100	100 / 100 / 100	90 / 100 / 90
Laya	37 / 0 / 0	27 / 0 / 0	20 / 0 / 0	28 / 0 / 0
Decider	73 / 73 / 53	100 / 0 / 0	90 / 22 / 20	88 / 28 / 24
SemIf-2B	100 / 13 / 13	100 / 0 / 0	93 / 25 / 23	98 / 12 / 12
SemIf-4B	60 / 94 / 57	97 / 86 / 83	77 / 70 / 53	78 / 83 / 64
Bespoke-Nimble	20 / 100 / 20	100 / 97 / 97	63 / 37 / 23	61 / 76 / 47
Luna Decisions	37 / 100 / 37	100 / 90 / 90	90 / 96 / 87	76 / 94 / 71
CoPE	83 / 84 / 70	100 / 50 / 50	100 / 53 / 53	94 / 61 / 58
BART	100 / 0 / 0	100 / 0 / 0	100 / 0 / 0	100 / 0 / 0
GPT-5.6 Luna	77 / 100 / 77	100 / 93 / 93	100 / 80 / 80	92 / 90 / 83
GPT-6 Sol	93 / 100 / 93	100 / 100 / 100	100 / 100 / 100	98 / 100 / 98

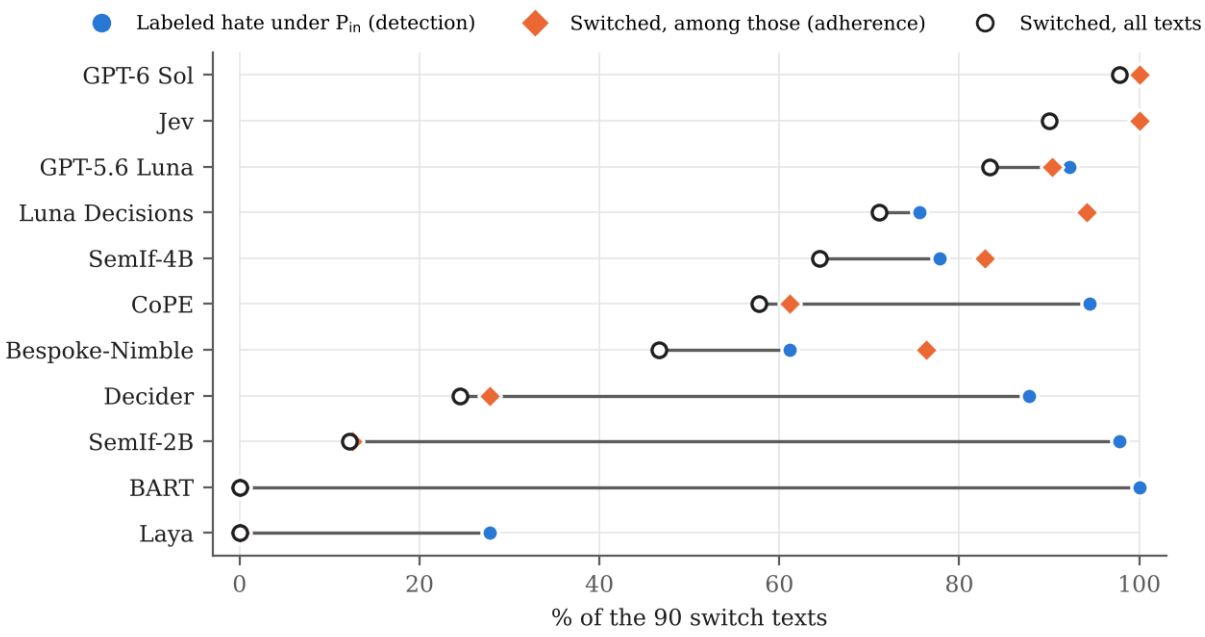


Figure S5. Detection and adherence on the 90 switch texts. The gap between detection and the switch among all texts shows texts that a model labels hate under P_{in} but fails to switch.

What this shows. The policy contrast separates two failure modes. Jev, Sol and Luna Decisions switch 94% to 100% of the texts they label hate under P_{in}, and Luna 90%, so their misses come mostly from not labeling hate against men (P_{in} accuracy 70% for Jev and 37% for Luna Decisions). Decider, SemIf-2B and BART label almost every switch text hate but rarely switch, so they ignore the policy change. Laya fails at both. Decider switches 73% of the texts about men it labels hate but none about white people, so adherence also varies by target. Accepting only the 234 validated texts changes switch rates by −1.5 to +6.4 points and leaves the ranking unchanged.

S13 Decider Training Exposure

Replaying Decider's public training code on the MHS corpus assigns our MHS test comments to three partitions (Table S13a). The replay draws from 135,556 rows and 39,565 eligible comments, capping training at 20,000 comments.

Table S13a. Partitions of our MHS test set under the replayed training selection.

Partition of our MHS test set	Comments	Hate	Non-hate
Replayed training selection	1603	403	1200
Replayed evaluation partition	123	29	94
Neither	1477	350	1127

Table S13b. Benchmark-direct macro-F1 on the 1,477 comments in neither partition. TF-IDF+LR and DistilBERT are trained on MHS.

Model	Macro-F1	95% CI
Decider	.780	[.756, .803]
TF-IDF+LR	.785	[.759, .811]
DistilBERT	.809	[.783, .833]
Jev	.691	[.664, .717]
GPT-6 Sol	.692	[.665, .720]

Table S13c. Decider against the supervised classifiers on the same 1,477 comments.

Contrast	Δ (pp)	Unadjusted 95% CI	Adjusted 95% CI (2 contrasts)
Decider − TF-IDF+LR	−0.6	[−3.4, +2.2]	[−3.8, +2.7]
Decider − DistilBERT	−2.9	[−5.5, −0.3]	[−5.8, +0.1]

What this shows. On comments outside the replayed selection, Decider still scores about 9 points above Jev and Sol but no better than classifiers trained on MHS. Its MHS lead is therefore in-domain training, not general ability.

S14 Additional Annotators

To add a guardrail model to the annotation reference, we ran GPT-OSS Safeguard 20B and NVIDIA Nemotron Content Safety through OpenRouter on the audit and policy-contrast tasks on October 21, 2026.

Table S14. Attempted annotation runs. Recorded charges exclude reserved amounts for requests without a final charge.

Model	Task	Requested	Attempted	Valid	Failed or invalid	Recorded $	Outcome
GPT-OSS Safeguard 20B	Audit	200	15	12	3	0.0021	Rate limits, incomplete
GPT-OSS Safeguard 20B	Policy contrast	720	44	42	2	0.0038	Rate limits, incomplete
Nemotron Content Safety	Pilot, both tasks	6	6	0	6	0.0012	Invalid format, full run not started

Safeguard returned 54 valid annotations, but repeated rate-limit errors stopped both tasks, including after a paced continuation and a bounded delay probe. Nemotron returned its native safe or unsafe ratings rather than the requested structured annotations, so all six pilot responses failed schema validation.

What this shows. Neither model produced a complete annotation reference, so the audit keeps the expert and two LLM judges. These are execution failures, not evidence about either model's accuracy on hate-speech.

S15 Compute and Hosted Spend

Local models ran on one NVIDIA T4 GPU (16 GB) with two Intel Xeon Gold 6230 processors (40 cores, 80 threads) and 96 GB RAM, using torch 2.11 with CUDA 12.8. Decider, SemIf and BART ran in BF16, and Bespoke-Nimble and CoPE in NF4 with BF16 compute.

Table S15. Recorded hosted charges by experiment (USD). Ledgers were kept per experiment, so amounts are not summed across rows.

Experiment	Jev	Luna Decisions	Luna	Sol
Main experiments, catalogued	2.66		2.47	28.35
Main experiments, later increment	0.12		1.84	3.65
All Luna Decisions runs		3.98		
Sol without definition				4.30
Definition paraphrases	0.12	0.16	0.35	3.35
B* validation	0.034	0.073		

What this shows. Hosted charges stay in the tens of dollars, and Sol accounts for most of them. Its benchmark-direct price per text is about ten times Luna's and thirty times Jev's.

What the Supplement Adds

- **Rankings depend on the prompt.** No prompt condition gives a stable model ranking (S1), and even a generic definition moves macro-F1 by up to 23.6 points (S2).
- **Definitions change more than macro-F1 shows.** Corrected and broken predictions offset each other (S5), the definition rather than the class descriptions drives the effect (S6), and option names matter more than definition wording (S7).
- **Offensive language remains the hardest boundary.** Decision models label up to 78% of offensive posts hate (S3).
- **Decomposition is an operating point, not an accuracy gain.** Threshold tuning recovers most of its gains, soft aggregation does not help, and many corrected verdicts rest on wrong components (S8).
- **Cost, not latency, separates the deployable options.** Decomposition raises hosted cost even where latency is flat (S9), and cascades save cost but never beat their stronger model (S10).
- **The audit does not hinge on the LLM judges.** The expert alone gives the same ranking and conflict rates (S11).
- **Policy failures have two causes.** Some models miss the target group, others ignore the policy change (S12).
- **Decider's MHS lead is in-domain training** (S13).